Practical No : 17

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Q.17) Write a C program for travelling salesman problem.

```
#include <stdio.h>
#include <limits.h>

#define MAX 10

int n;
int cost[MAX][MAX];
int visited[MAX];
int bestCost = INT_MAX;
int bestPath[MAX + 1];
int path[MAX + 1];

void tsp(int city, int count, int currentCost)
{
   int i;

   // All cities have been visited
   if (count == n) {
      // Return to the starting city
      int totalCost = currentCost + cost[city][0];

      if (totalCost < bestCost) {
         bestCost = totalCost;

         for (i = 0; i < n; i++)
            bestPath[i] = path[i];
```

```
            bestPath[n] = 0;
        }

        return;
    }

    // Try visiting every unvisited city
    for (i = 0; i < n; i++) {
        if (!visited[i]) {
            visited[i] = 1;
            path[count] = i;

            tsp(i, count + 1, currentCost + cost[city][i]);

            visited[i] = 0;
        }
    }
}

int main()
{
    int i, j;

    printf("Enter number of cities: ");
    scanf("%d", &n);

    printf("Enter the cost matrix:\n");
    for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
            scanf("%d", &cost[i][j]);
        }
```

```
    }

    // Start from city 0
    for (i = 0; i < n; i++)
        visited[i] = 0;

    visited[0] = 1;
    path[0] = 0;

    tsp(0, 1, 0);

    printf("\nMinimum cost: %d\n", bestCost);

    printf("Optimal path: ");
    for (i = 0; i <= n; i++) {
        printf("%d", bestPath[i] + 1);
        if (i < n)
            printf(" -> ");
    }

    printf("\n");

    return 0;
}
```

Output:

